

Haris Kamal Rana

Computer Science Undergraduate
Building Intelligent Systems for the Physical World

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EDUCATION

The University of Lahore

Bachelor of Science in Computer Science (BSCS), CGPA: 3.66/4.00

Lahore, Pakistan

Sep. 2023 – Expected Jun. 2027

ENGINEERING EXPERIENCE

DriveVitals – Founder & Lead Software Engineer | [GitHub](#)

Jul. 2026 – Present

Independent Engineering Project

Remote

- Leading the development of **DriveVitals**, a Digital Twin and fleet intelligence platform that simulates commercial fleet operations, vehicle telemetry, driver behaviour, and real-time fleet monitoring.
- Architected a modular backend using **Python, FastAPI, WebSockets, and PostgreSQL**, enabling real-time telemetry streaming, fleet management, analytics services, and a React-based monitoring dashboard.
- Developed the core Digital Twin runtime, implementing simulation orchestration, vehicle control, driver decision-making, physics-based vehicle behaviour, telemetry generation, and domain-driven backend services.
- Designed a scalable software architecture following **SOLID principles**, dependency injection, and modular component design to support future integration of OBD-II telemetry and machine learning capabilities.
- Leading a team of three developers by defining project milestones, managing feature-based Git workflows, reviewing pull requests, and coordinating controlled integration into the development branch.
- Designed the relational data model for fleet operations, historical telemetry, trip management, driver performance, and vehicle health analytics while maintaining extensibility for future predictive maintenance features.

SELECTED ENGINEERING PROJECTS

AI Knowledge Platform | [GitHub](#) | *Python, Streamlit, Groq API, RAG, Vector Search*

2026

- Built an AI-powered knowledge platform that answers engineering questions from a personal Obsidian knowledge base using Retrieval-Augmented Generation (RAG).
- Developed an indexing pipeline with document chunking, embeddings, and vector search to provide context-aware responses through the Groq API.
- Implemented an automated GitHub commit watcher that converts development activity into structured engineering notes and synchronizes them with an Obsidian vault.
- Designed a modular architecture separating document ingestion, indexing, retrieval, and response generation for future multi-model extensibility.

Smart Door Security System | [GitHub](#) | *Python, OpenCV, Raspberry Pi, Computer Vision*

2025

- Built an embedded access-control system using Python, OpenCV, and Raspberry Pi for real-time facial authentication.
- Implemented a computer vision pipeline combining face detection, feature encoding, facial recognition, and access decision logic.
- Developed head pose-based liveness verification with randomized prompts to reduce spoofing attacks and improve authentication reliability.
- Integrated GPIO-based door control with PIN fallback authentication and event logging for secure physical access.

Vehicle Service Workshop Management Database | [GitHub](#) | *SQL, MySQL, Database Design*

2026

- Designed a normalized relational database for managing customers, vehicles, repair orders, mechanics, inventory, and payment records.
- Applied normalization (1NF–3NF), foreign-key constraints, and indexing strategies to ensure data integrity and optimize query performance.
- Developed SQL queries, views, and reports supporting service history, workshop operations, and business analytics.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JavaScript, TypeScript, Java
Frameworks & Libraries: FastAPI, React, Streamlit, OpenCV, SQLAlchemy, Pydantic, NumPy, Pandas, Matplotlib
AI & Intelligent Systems: Retrieval-Augmented Generation (RAG), LLM Integration, Vector Search, Computer Vision, Digital Twins, Vehicle Telemetry
Databases: PostgreSQL, MySQL, Database Design, Relational Modeling
Software Engineering: REST APIs, WebSockets, Object-Oriented Programming, SOLID Principles, Design Patterns, Git, GitHub, Agile Development
Developer Tools: Docker, Linux, VS Code, Postman, Raspberry Pi

CERTIFICATIONS

Elements of AI	Jul. 2025
<i>University of Helsinki & MinnaLearn</i>	<u>Credential</u>